

OLIST

Data Analysis Project

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Client/Sponsor: Olist Operation Manager

Purpose:

This project is being conducted to investigate customer dissatisfaction in the OLIST e-commerce dataset. The stakeholder reported a possible connection between delivery performance, post-delivery customer experience, and negative customer reviews, but no predefined KPIs or structured analysis were available.

The purpose of the project is to translate this business problem into measurable analytical outputs, define delivery and customer satisfaction KPIs, and evaluate whether delivery performance is associated with customer dissatisfaction.

The analysis focuses on delivered orders linked to valid customer reviews and examines:

- actual delivery performance;
- estimated versus actual delivery experience;
- delivery delay and expectation gap;
- review score and negative review rate;
- operational patterns that may help explain negative customer experiences.

After the initial Scope of Work review, the analysis was extended to include delivery expectation gap metrics, comparing estimated delivery dates with actual delivery dates and evaluating their relationship with review scores.

After the interim stakeholder review, two additional follow-up questions were added before final recommendations:

1. whether the concentration observed in São Paulo seller areas reflects a true performance issue or mainly a volume effect;

2. whether customer dissatisfaction is driven by absolute delay or by the gap between expected and actual delivery.

The final goal is to identify relevant associations, validate the strongest operational patterns, clarify the limits of the available data, and define which additional operational data is required before proposing final recommendations.

Scope / Major Project Activities:

What are the major parts of this project? List out the high-level steps, activities, or stages of the project, and give a brief description for each.

Activity	Description
Data understanding and validation	<i>Review the available OLIST tables, confirm data structure, validate data types, and identify any data quality issues that may affect the analysis.</i>
KPI definition and readiness check	<i>Translate the stakeholder's initial business problem into measurable KPIs related to delivery performance, customer reviews, delivery expectation gap, and their relationship. Verify that the required KPIs are computable.</i>
Delivery performance and expectation gap analysis	<i>Calculate actual delivery time, estimated delivery time, delivery delay, delivery expectation gap, late delivery flag, and late delivery rate to understand both operational delivery performance and the gap between promised and actual delivery experience.</i>
Customer review analysis	<i>Analyze review scores and negative review rate to measure customer satisfaction across delivered orders.</i>

<i>Delivery expectation vs review analysis</i>	<i>Compare review scores and negative review rates across on-time deliveries, late deliveries, early deliveries, and different delivery expectation gap ranges to evaluate whether customer dissatisfaction is linked only to delays or also to expectation management.</i>
<i>Interim stakeholder review</i>	<i>Share preliminary numerical findings with the stakeholder to confirm whether the analysis direction is aligned with the business question before completing exploratory segmentation and final recommendations.</i>
<i>Post-stakeholder follow-up analysis</i>	<i>Compare São Paulo seller areas against the rest of the dataset to validate whether the observed concentration is driven by volume or worse performance. Complete expectation gap analysis against review score and negative review rate before moving to exploratory segmentation and final recommendations.</i>
<i>Exploratory segmentation analysis</i>	<i>Explore secondary patterns by customer state, product category, seller, freight value, and other available operational dimensions.</i>
<i>Insights and recommendations</i>	<i>Summarize key findings, limitations, and practical recommendations for the stakeholder.</i>

Deliverables:

A specific list of things that your project will deliver.

Deliverable	Description / Details
KPI readiness summary	A summary confirming that the selected KPIs are computable and that the dataset is ready for analysis.
Delivery performance KPI table	A table containing actual delivery time, estimated delivery time,

	delivery delay, delivery expectation gap, late delivery flag, and late delivery rate.
Customer review KPI table	A table containing average review score and negative review rate.
Late vs on-time delivery comparison	A comparison of review scores and negative review rate between late, on-time, and early deliveries.
Delivery expectation gap analysis	A table and interpretation showing the difference between estimated delivery time and actual delivery time, and how this gap relates to review scores and negative review rate.
Interim stakeholder review summary	A short interim summary of preliminary delivery, review, and expectation gap findings shared with the stakeholder to confirm analytical direction before completing exploratory segmentation and final recommendations.
Post-stakeholder follow-up summary	A short summary comparing São Paulo seller areas against the rest of the dataset and completing the delivery expectation gap vs review analysis.
Exploratory segmentation outputs	Analysis by customer state, product category, seller, freight value, and other relevant operational dimensions available in the dataset.
Key insights summary	A concise written summary of the main findings, business interpretation, and relevant limitations.
Operational recommendations	Practical recommendations based on the analysis results, without making final business policy decisions.

Schedule Overview / Major Milestones:

The expected schedule for the project. This can be defined by milestones (e.g. "all data is cleaned and processed"), periods of time ("Week 1 / Week 2"), or other ways based on the needs of the project.

Milestone	Expected Completion Date	Description / Details
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<i>Delivery performance and expectation gap analysis completed</i>	<i>April 28, 2026</i>	<i>Actual delivery time, estimated delivery time, delivery delay, delivery expectation gap, late delivery flag, and late delivery rate calculated and reviewed.</i>
<i>Customer review analysis completed</i>	<i>April 28, 2026</i>	<i>Average review score and negative review rate calculated and reviewed.</i>
<i>Delivery expectation vs review comparison completed</i>	<i>April 29, 2026</i>	<i>Review scores and negative review rates compared across late, on-time, early deliveries, and delivery expectation gap ranges.</i>
<i>Stakeholder interim review completed</i>	<i>April 29, 2026</i>	<i>Preliminary numerical findings shared with the stakeholder to confirm whether the analysis direction is aligned with the business question. Feedback collected before proceeding with exploratory segmentation and final recommendations.</i>
<i>Post-stakeholder follow-up analysis completed</i>	<i>April 30, 2026</i>	<i>São Paulo vs rest-of-dataset comparison completed, including order volume, late delivery rate, negative review rate, average review score, and approval-to-carrier timing. Delivery expectation gap vs review analysis completed.</i>
<i>Exploratory segmentation completed</i>	<i>April 30, 2026</i>	<i>Secondary patterns reviewed by customer state, product category, seller, freight value, and other relevant operational dimensions.</i>
<i>Final insights and recommendations delivered</i>	<i>May 4, 2026</i>	<i>Key findings, limitations, visual outputs, and operational recommendations summarized for stakeholder review.</i>

*Estimated date for completion:

This is my "if all goes well and I have everything I need, this is when I'll be done" date.

*Estimated date for completion:

May 4, 2026

This is the expected completion date if all required data remains available and the post-stakeholder follow-up analysis does not require a major change in analytical direction.